

Conjecture 00000001937 is false:
the subgroup zeta function of $U(\mathbb{Z}) \cong \mathbb{Z}$ has abscissa 1, not
 $\dim / (\dim + 1) = 1/2$

Jackmeson1

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1 The conjecture

Conjecture 00000001937 (English text, verbatim): “Definition: The subgroup growth zeta function is the Dirichlet series of subgroup-index counts. Conjecture: The abscissa of convergence of the zeta function of an arithmetic linear group is $\dim G / (\dim G + 1)$ (an explicit rational), uniquely determined by the algebraic dimension of the group.” The Chinese text says the same.

Reading. For a group G let $a_n(G)$ be the number of subgroups of G of index n , and let $\zeta_G(s) = \sum_{n \geq 1} a_n(G) n^{-s}$ (the “Dirichlet series of subgroup-index counts”). The conjecture asserts that for every arithmetic linear group G (integer points of a linear algebraic group $\mathbf{G}$ over $\mathbb{Q}$) the abscissa of convergence of ζ_G equals $\dim \mathbf{G} / (\dim \mathbf{G} + 1)$. We give one arithmetic group for which this fails, so the universally quantified statement is false; the second clause (“uniquely determined by the dimension”) is part of the same conjunction and needs no separate treatment.

Arithmetic groups. We use the definition from Wikipedia, “Arithmetic group”: “If $\mathbf{G}$ is an algebraic subgroup of $\mathrm{GL}_n(\mathbb{Q})$ for some n then we can define an arithmetic subgroup of $\mathbf{G}(\mathbb{Q})$ as the group of integer points $\Gamma = \mathrm{GL}_n(\mathbb{Z}) \cap \mathbf{G}(\mathbb{Q})$.” Take $\mathbf{G} = U_2$, the algebraic subgroup of GL_2 of upper unitriangular matrices $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ (defined by $x_{11} = x_{22} = 1$, $x_{21} = 0$). It is isomorphic to the additive group, of dimension 1. Its arithmetic subgroup is

$$U(\mathbb{Z}) = \mathrm{GL}_2(\mathbb{Z}) \cap U_2(\mathbb{Q}) = \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\} \cong \mathbb{Z}.$$

The conjecture predicts abscissa $1/(1+1) = 1/2$ for $U(\mathbb{Z})$.

2 Definitions (as in the Lean file)

- `subgroupCount G n` = $a_n(G)$ is the cardinality (`Nat.card`) of $\{H \leq G : [G : H] = n\}$, with Mathlib’s `Subgroup.index`. Mathlib’s convention is $[G : H] = 0$ for infinite index, so a_0 counts infinite-index subgroups; it does not enter the Dirichlet series (Mathlib’s `LSeries.term f s 0 = 0`).
- `subgroupZeta G s` = `LSeries (n ↦ a_n(G)) (s)` = $\sum_{n \geq 1} a_n(G) n^{-s}$.

- `subgroupZetaAbscissaAbs G` = Mathlib's `LSeries.abscissaOfAbsConv`: the infimum in $\overline{\mathbb{R}}$ of the real x at which the series converges absolutely.
- `subgroupZetaAbscissa G`: the infimum in $\overline{\mathbb{R}}$ of the real σ for which the ordered partial sums $\sum_{k=1}^N a_k(G)k^{-\sigma}$ have a finite limit as $N \rightarrow \infty$ (the abscissa of convergence, read at real points).
- `U`: the range of the homomorphism $n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ from $\mathbb{Z}$ to $\mathrm{SL}_2(\mathbb{Z})$; `mem_U_iff` proves that U is exactly the set of matrices in $\mathrm{SL}_2(\mathbb{Z})$ with $x_{11} = x_{22} = 1$, $x_{21} = 0$, i.e. $\mathrm{GL}_2(\mathbb{Z}) \cap U_2(\mathbb{Q})$ (such a matrix has determinant 1 automatically).

3 Proof

Lemma 1 (`subgroupCount_U`). $a_n(U(\mathbb{Z})) = 1$ for every $n \in \mathbb{N}$.

Proof. $n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ is an injective homomorphism, so $U(\mathbb{Z}) \cong \mathbb{Z}$ and $a_n(U(\mathbb{Z})) = a_n(\mathbb{Z})$ (`subgroupCount_eq_add`: isomorphisms preserve indices). Every subgroup of $\mathbb{Z}$ is cyclic, $H = a\mathbb{Z} = |a|\mathbb{Z}$, and $[\mathbb{Z} : |a|\mathbb{Z}] = |a|$ (Mathlib: `Int.subgroup_cyclic`, `Int.index_zmultiples`). Hence $H = [\mathbb{Z} : H]\mathbb{Z}$, and the only subgroup of index n is $n\mathbb{Z}$. $\square$

Theorem 2. $\zeta_{U(\mathbb{Z})}(s) = \zeta(s)$ (Riemann zeta) for $\mathrm{Re} s > 1$, and both abscissae of $\zeta_{U(\mathbb{Z})}$ equal 1.

Proof. By the lemma the coefficient sequence is the constant 1. Mathlib gives `LSeries_one_eq_riemannZeta` and `LSeries.abscissaOfAbsConv_one`. At a real σ the partial sums $\sum_{k \leq N} k^{-\sigma}$ have nonnegative terms, so they have a finite limit iff the series is summable, iff $\sigma > 1$ (`Real.summable_one_div_nat_rpow`); hence the set of real convergence points is the same as Mathlib's set for absolute convergence, and `subgroupZetaAbscissa U = 1`. $\square$

Theorem 3 (`conjecture_1937_false`). For every $d \in \mathbb{N}$, $d/(d+1) < 1$, so neither abscissa of $\zeta_{U(\mathbb{Z})}$ equals $d/(d+1)$; in particular neither equals $1/2 = \dim U_2/(\dim U_2 + 1)$. The conjecture is false.

Since the conclusion holds for every natural number d , it does not depend on how the dimension of the ambient algebraic group is computed; the Lean file does not formalize algebraic dimension.

4 Lean formalization

File `lean/Conjecture1937/Basic.lean` (Lean 4, Mathlib v4.33.1, 206 lines, only `import Mathlib`). Main theorem `C1937.conjecture_1937_false` states, as one conjunction: `mem_U_iff`; $\forall n$, `subgroupCount U n = 1`; $\forall s$, $1 < \mathrm{Re} s \rightarrow \text{subgroupZeta } U \text{ } s = \text{riemannZeta } s$; `subgroupZetaAbscissa U = 1`; `subgroupZetaAbscissaAbs U = 1`; for every $d : \mathbb{N}$ both abscissae differ from $((d : \mathbb{R})/(d+1) : \mathbb{R})$; and `subgroupZetaAbscissa U  $\neq$  1/2`. Supporting theorems: `eq_zmultiples_index`, `subgroupCount_eq_add`, `partialSum_U_converges_iff`, `subgroupZetaAbscissa_U`, `subgroupZetaAbscissaAbs_U`, `formula_ne_one`. `lake build` succeeds; `#print axioms C1937.conjecture_1937_false` gives `[propxt, Classical.choice, Quot.sound]`; no sorry, no native_decide.

Scope. Refuted: the formula for the arithmetic group $U(\mathbb{Z})$ of the unipotent group $U_2 \cong \mathbb{G}_a$, with the abscissa read as absolute convergence or as convergence of the ordered partial sums at real points. Not covered: a reading in which “arithmetic linear group” is restricted to arithmetic subgroups of semisimple groups (e.g. $\mathrm{SL}_n(\mathbb{Z})$); no such group is treated here. The real-point definition of the abscissa of (ordinary) convergence is used; conditional convergence at non-real s is not formalized.

5 Relation to the earlier submission

PR #264 (orionsheep) used the same counterexample ($G = \mathbb{Z}$, subgroups $n\mathbb{Z}$, $\zeta_{\mathbb{Z}}$ the Riemann zeta function). It was closed because “The theorem proves $1 \neq 2 \wedge 25 > 24 \wedge 2 = 2 \cdot 1$; the entire $G = \mathbb{Z}$ counterexample ... lives in prose ... Nothing about Dirichlet series or subgroup zeta functions is formalized.” This submission defines the subgroup counts $a_n(G)$ for an arbitrary group with Mathlib’s index, the subgroup zeta function as a Mathlib `LSeries`, and both abscissae; it constructs $U(\mathbb{Z})$ as a subgroup of $\mathrm{SL}_2(\mathbb{Z})$, proves that it has exactly one subgroup of each index, and proves that its abscissae equal 1, which differs from $d/(d+1)$ for every d .

Source

Wikipedia, “Arithmetic group”, https://en.wikipedia.org/wiki/Arithmetic_group (definition of arithmetic subgroup quoted above). Mathlib (lemmas named above).